TUTORIAL - 9: Black-Box & White-Box Testing

Aim

Analyze Black-Box Testing, White-Box Testing, Integration Testing, and System Testing on a Sample Web Application.

Description

Software testing ensures that an application works correctly and meets user requirements. Different testing techniques are used depending on the testing objective:

- **Black-Box Testing:** A testing technique where the tester verifies the application's functionality without knowing the internal source code.
- **White-Box Testing:** A testing technique where the tester has complete knowledge of the application's internal source code to verify program logic, conditions, loops, branches, and statements.
- **Integration Testing:** Verifies whether multiple modules work together correctly.
- **System Testing:** Verifies the complete application end-to-end.

Problem Statement

Perform Black-Box Testing, White-Box Testing, Integration Testing, and System Testing on the **Login Module** of the Student Management System. Execute test cases, analyze the results, and document the testing activities.

Implementation Details

Parameter	Details
Project Name	Student Management System
Application / Software	Student Management System

Parameter	Details
Module Name	Login Module
SDLC Model	Waterfall Model
Programming Language	Python
IDE / Tool Used	Visual Studio Code / PyCharm
Input / Test Data	admin/admin123, admin/12345, st…

Source Code

```
username = input("Enter Username: ")

password = input("Enter Password: ")

if username == "admin" and password == "admin123":

    print("Login Successful")

    print("Dashboard Opened")

else:

    print("Invalid Username or Password")
```

Detailed Test Case (TC001)

Attribute	Details	Attribute	Details
Requirement ID	R001	**Test Environment**	Windows 10
Feature Under Test	User Authentication	**Browser / OS**	Google Chrome
Test Scenario	Login with valid username and password	**Test Data**	admin / admin123

Attribute	Details	Attribute	Details
Test Objective	Verify successful login	**Expected Result**	Login Successful and Dashboard Opens
Test Type	System Testing	**Actual Result**	Login Successful and Dashboard Opens
Test Technique	Black Box Testing	**Status**	Pass
Priority / Severity	High / Critical	**Executed By**	Student Name
Preconditions	Login page should be displayed	**Execution Date**	22/10/2026
Postconditions	Dashboard should open		

Test Case Execution Steps

Step	Action	Input Data	Expected Result	Actual Result	Status
1	Launch Application	-	Login Page displayed	Login Page displayed	Pass
2	Enter Username	admin	Username accepted	Username accepted	Pass
3	Enter Password	admin123	Password accepted	Password accepted	Pass
4	Click Login	Click	Dashboard Opens	Dashboard Opens	Pass
5	Logout	Click	Login Page Opens	Login Page Opens	Pass

Testing Activities Performed

Lifecycle Phase	Activity Phase	Activity Details	Deliverable
Requirement Analysis	Analysis Phase	Analyze Login Requirements	SRS Document
Design	Test Planning	Prepare Test Plan	Test Plan

Lifecycle Phase	Activity Phase	Activity Details	Deliverable
Development	Test Case Development	Create Black-Box & White-Box Test Cases	Test Case Document
Testing	Test Execution	Execute Black-Box, White-Box, Integration & System Tests	Test Execution Report
Testing	Test Closure	Prepare Test Summary	Test Summary Report

Observations

1. **Black-Box Testing:** Completed successfully — **Pass**
2. **White-Box Testing:** Branch coverage verified — **Pass**
3. **Integration Testing:** Login communicates correctly with Dashboard — **Pass**
4. **System Testing:** Complete application tested successfully — **Pass**
5. **Defect Status:** No major defects found — **Completed**

Console Output

Student Management System

LOGIN

Username: admin

Password: admin123

Login Successful

Dashboard Opened

Black-Box Testing : PASS

White-Box Testing : PASS

Integration Testing : PASS

System Testing : PASS

Overall Result: Testing Completed Successfully

Questionnaire Solutions

1. Which testing technique verifies functionality without examining source code?

- **Answer:** C) Black-Box Testing

2. Which testing technique requires knowledge of internal program code?

- **Answer:** D) White-Box Testing

3. Integration Testing is mainly used to verify:

- **Answer:** B) Communication between modules

4. System Testing validates:

- **Answer:** C) Complete Application Behavior

5. Which testing checks execution paths and branch coverage?

- **Answer:** B) White-Box Testing

Conclusion & Outcomes

- Understood the concepts and practical application of Black-Box Testing, White-Box Testing, Integration Testing, and System Testing.
- Successfully designed and executed test cases using both Black-Box and White-Box testing techniques.
- Verified seamless communication between the Login and Dashboard modules through Integration Testing.
- Validated the end-to-end functionality of the Student Management System using System Testing.
- Learned how structured, multi-level testing techniques ensure software quality, reliability, and correctness prior to deployment.